

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : E & TC Semester : III Subject Code & Name: BTES304 Electrical Machines and Instruments Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	With suitable diagram explain construction of DC machine.		06
B)	With the help of neat diagram explain three point starter of DC machine.		06
C)	List various speed control methods of DC motors. Explain any one.		06
Q.2	Solve Any Two of the following.		12
A)	Explain autotransformer starter for induction motor with the help of suitable diagram.		06
B)	What is hunting? Explain causes, effect of hunting in synchronous motors.		06
C)	Explain working principle of synchronous motors with suitable diagram.		06
Q. 3	Solve Any Two of the following.		12
A)	Explain working of stepper motor with the help of neat diagram.		06
B)	Explain working of Linear induction motor with the help of neat diagram.		06
C)	Explain working of FHP motor with the help of neat diagram.		06
Q.4	Solve Any Two of the following.		12
A)	With the help of suitable diagram explain working of LVDT.		06
B)	List different temperature transducers. Explain any one.		06
C)	With the help of suitable diagram explain working of Hall effect transducer.		06
Q. 5	Solve Any Two of the following.		12
A)	With the help of suitable diagram explain working of sound level meter.		06
B)	With the help of block diagram explain working of X-Y plotter.		06
C)	Explain working of object counter with the help of neat diagram.		06
*** End ***			

The grid and the borders of the table will be hidden before final printing.